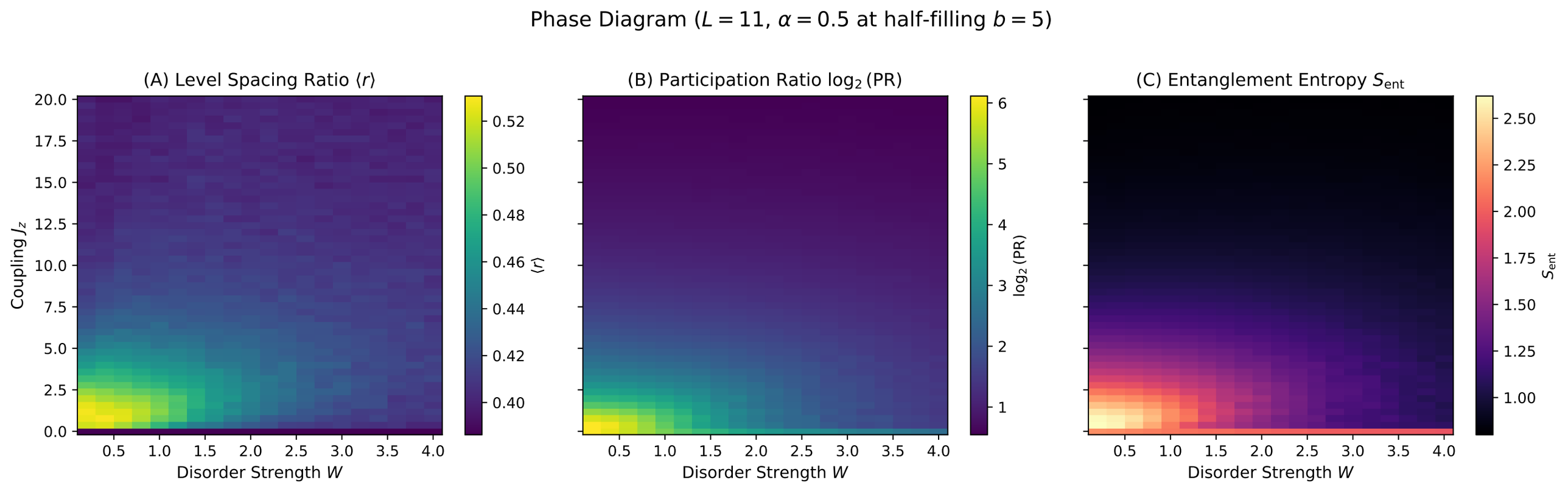

Phase Diagram ($L = 11$, $\alpha = 0.5$ at half-filling $b = 5$)
(A) Level Spacing Ratio $\langle r \rangle$
(B) Participation Ratio $\log_2(\mathrm{PR})$
(C) Entanglement Entropy S_{ent}
Coupling J_z
Disorder Strength W
$\langle r \rangle$
$\log_2(\mathrm{PR})$
S_{ent}